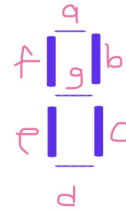


Karnaugh map

BCD:

A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	0	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	1	1	1
0	1	1	0	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	0	1	1	1
1	0	1	0	1	0	1	1	1	1	1
1	0	1	1	0	0	0	1	1	1	0
1	1	0	0	1	1	1	1	0	1	1
1	1	0	1	0	1	1	1	1	1	1
1	1	1	0	1	0	1	1	1	1	1
1	1	1	1	1	0	0	0	1	1	1

BCD to 7 segment



Not care: 10-15

a:

AB	CD	00	01	11	10
00		1		1	1
01			1	1	1
11		x	x	x	x
10		1	1	x	x

$$= B'D' + BD + C + A$$

$$= A + C + (B \oplus D)$$

b:

AB	CD	00	01	11	10
00		1		1	1
01		1		1	1
11		x	x	x	x
10		1	1	x	x

$$b: C'D' + CD + B'$$

$$; B' + (C \oplus D)$$

c:

AB	CD	00	01	11	10
00		1	1	1	
01		1	1	1	1
11		x	x	x	x
10		1	1	x	x

$$c: C' + B + D$$

d:

AB	CD	00	01	11	10
00		1		1	1
01			1		1
11		x	x	x	x
10		1	1	x	x

$$d: A + CD' + B'D'$$

$$BC'D + B'C$$

e:

AB	CD	00	01	11	10
00		1			1
01					1
11		x	x	x	x
10		1		x	1

$$e: CD' + B'D'$$

f:

AB	CD	00	01	11	10
00		1			
01		1	1		1
11		x	x	x	x
10		1	1	x	x

$$f: BC' + C'D' + BD' + A$$

g:

AB	CD	00	01	11	10
00				1	1
01		1	1		1
11		x	x	x	x
10		1	1	x	x

$$g: A + BC' + CD' + B'C$$